

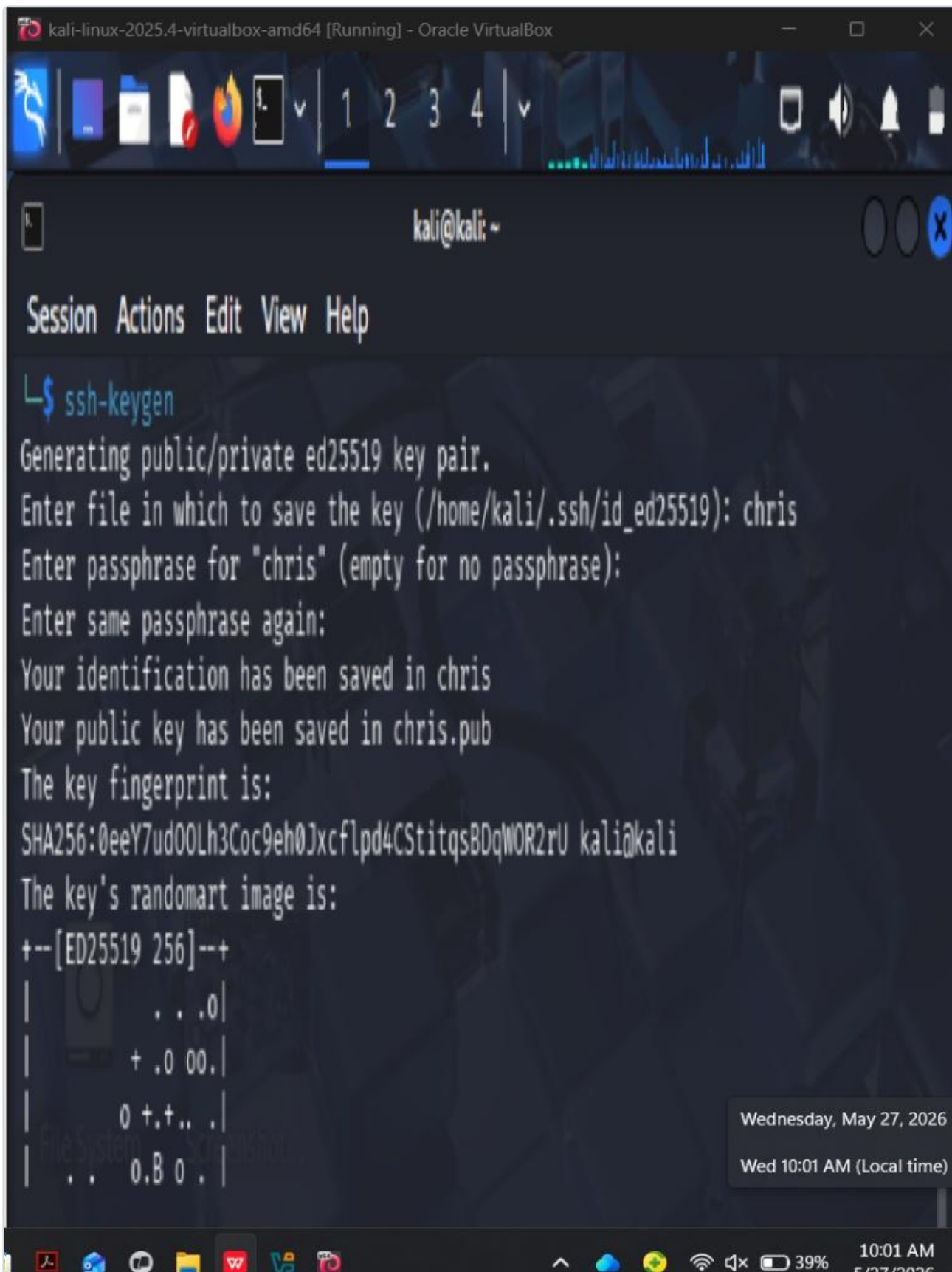
Cryptographic Key Management: Ed25519

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Key Pair Generation Walkthrough

Using the standard Linux terminal interface utility, a customized secure key pair can be crafted using the following interactive sequence:

```
ssh-keygen -t ed25519
```



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
kali@kali: ~
Session Actions Edit View Help
$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/kali/.ssh/id_ed25519): chris
Enter passphrase for "chris" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in chris
Your public key has been saved in chris.pub
The key fingerprint is:
SHA256:0eeY7ud00Lh3Coc9eh0Jxcflpd4CStitqsBDqWOR2rU kali@kali
The key's randomart image is:
+--[ED25519 256]--+
|      . . .0|
|      + .0 00.|
|      0 +.+.. .|
|      . . 0.B 0 .|
```

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Figure 1: Terminal execution detailing identity block creation, key saving targets ('chris'), and fingerprint randomart matrix outputs.

Key Structural Manifest

The Ed25519 algorithm produces two components: the private identity token (which remains confidential) and the public distribution token (uploaded to remote servers). The graphical randomart helps visually verify the key integrity quickly.